Banking Sector Risk, Performance & Portfolio Intelligence in North America

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1. Introduction

The banking sector plays a central role in financial markets, acting as both a transmission mechanism for macroeconomic policy and a key driver of capital allocation. As such, bank equities are highly sensitive to economic conditions, interest rate movements, and financial market cycles.

This project develops a comprehensive analytical framework to evaluate the performance, risk characteristics, and macroeconomic sensitivities of major North American banks. The analysis focuses on leading Canadian banks: Royal Bank of Canada (RY.TO), Toronto-Dominion Bank (TD.TO), Bank of Montreal (BMO.TO), Bank of Nova Scotia (BNS.TO), and Canadian Imperial Bank of Commerce (CM.TO), alongside major U.S. banks including JPMorgan Chase (JPM), Bank of America (BAC), Goldman Sachs (GS), and Morgan Stanley (MS). Benchmark indices such as the S&P 500 and TSX Composite are also included for comparative purposes.

The objective is to provide a data-driven assessment of:

- Relative performance and risk-adjusted returns
- Exposure to extreme downside risk
- Forward-looking return uncertainty
- Sensitivity to macroeconomic variables
- Optimal portfolio allocation strategies

2. Data and Methodology

2.1 Data Sources

The analysis uses:

- Historical price data from Yahoo Finance
- Macroeconomic indicators from FRED (Federal Reserve Economic Data)

The dataset spans from 2014 till 2026, capturing multiple market regimes, including:

- Post-financial crisis recovery
- Low interest rate environment
- COVID-19 market shock

- High inflation and tightening cycles

Daily data is used for return and risk calculations, while monthly data is used for macroeconomic modeling.

2.2 Analytical Framework

The project is structured into five core components:

Performance Analysis

Performance is evaluated using both absolute and risk-adjusted metrics:

- Compound Annual Growth Rate (CAGR)
- Annualized volatility
- Sharpe ratio (risk-adjusted return)
- Sortino ratio (downside risk-adjusted return)
- Maximum drawdown

These metrics provide a comprehensive view of both return generation and risk exposure.

Tail Risk Modeling

Traditional volatility measures do not fully capture extreme market events. Therefore, tail risk is assessed using:

- Value at Risk (VaR 99%)
- Expected Shortfall (ES 99%)

VaR estimates the threshold loss under extreme conditions, while Expected Shortfall measures the average loss beyond that threshold, making it a more robust indicator of crash risk.

Monte Carlo Simulation

To assess forward-looking uncertainty, Monte Carlo simulations are conducted:

- Simulating one-year return distributions

- Estimating probabilities of loss
- Capturing the range of potential outcomes

This approach provides insight into future risk scenarios, rather than relying solely on historical observations.

Macroeconomic Sensitivity

Bank performance is strongly linked to macroeconomic conditions. This is analyzed using:

- ElasticNet regression (linear, regularized model)
- Random Forest (non-linear model with feature importance)

Key macro variables include:

- Interest rates (Fed funds rate, BoC policy rate)
- Inflation (CPI)
- Unemployment
- Government bond yields

This allows for identification of the primary economic drivers of bank returns.

Portfolio Optimization

Portfolio construction is implemented using:

- Mean-variance optimization
- Efficient frontier analysis
- Maximum Sharpe ratio portfolio

This framework determines optimal asset allocation under risk-return trade-offs.

3. Results and Discussion

3.1 Performance Analysis

The results reveal a clear distinction between Canadian and U.S. banking equities.

U.S. banks such as JPMorgan, Goldman Sachs, and Morgan Stanley exhibit higher absolute returns over the sample period. This reflects their exposure to capital markets activities, investment banking revenues, and higher sensitivity to economic cycles.

However, these higher returns come with significantly greater volatility and larger drawdowns. In contrast, Canadian banks demonstrate more stable and consistent performance, with lower volatility and smoother growth trajectories.

Among all institutions analyzed, Royal Bank of Canada (RY.TO) emerges as the top performer on a risk-adjusted basis. This suggests that its returns are achieved with comparatively lower risk, making it a strong candidate for long-term investment.

Overall, the findings highlight a trade-off:

- U.S. banks: higher return potential, higher risk
- Canadian banks: lower return variability, stronger stability

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3.2 Tail Risk Analysis

Tail risk analysis provides deeper insight into extreme market behavior.

The results show that U.S. banks exhibit more pronounced left-tail risk, meaning they are more susceptible to large losses during market downturns. Institutions such as Bank of America and Morgan Stanley display the highest Expected Shortfall values, indicating substantial downside exposure in adverse scenarios.

Canadian banks, on the other hand, demonstrate lower tail risk, with smaller Expected Shortfall values and less severe drawdowns. This suggests a more resilient structure, likely driven by:

- Conservative regulatory frameworks
- Strong retail banking focus
- Lower reliance on volatile trading revenues

These findings reinforce the notion that Canadian banks provide defensive characteristics within the financial sector.

3.3 Monte Carlo Simulation

Monte Carlo simulations further emphasize the differences in risk profiles.

The simulated return distributions show that U.S. banks have wider dispersion, indicating higher uncertainty and more extreme potential outcomes. While this includes higher upside potential, it also introduces greater downside risk.

In contrast, Canadian banks display narrower distributions, reflecting more predictable return patterns and lower probability of loss.

This suggests that:

- U.S. banks are more suitable for growth-oriented strategies
- Canadian banks are better suited for risk-controlled portfolios

The simulation results align closely with historical risk observations.

3.4 Macroeconomic Sensitivity

The macroeconomic analysis reveals that bank equities are strongly influenced by:

- Interest rate movements
- Inflation dynamics
- Bond yields

Random Forest models identify these variables as the most important drivers of returns, highlighting the central role of monetary policy and inflation expectations.

ElasticNet results show limited significance for some banks, suggesting that the relationship between macro variables and returns may be non-linear and complex. This justifies the use of machine learning models to capture interactions not visible in linear frameworks.

From an economic perspective, this is intuitive:

- Higher interest rates can improve bank profitability through net interest margins
- Inflation affects lending demand and real returns
- Bond yields reflect broader financial conditions

3.5 Portfolio Optimization

Portfolio optimization demonstrates the benefits of diversification across North American banks.

The efficient frontier illustrates the trade-off between expected return and volatility, showing that optimal portfolios can achieve higher risk-adjusted returns through diversification.

The maximum Sharpe ratio portfolio tends to allocate a larger weight to Canadian banks, reflecting their stability and favorable risk-adjusted performance.

This suggests that combining:

- Canadian banks (stability)
- U.S. banks (growth)

can produce a well-balanced portfolio.

4. Conclusion

This project applies quantitative finance and data science techniques to evaluate banking sector equities in a structured and comprehensive manner.

Key Findings

- Canadian banks deliver strong risk-adjusted returns and lower volatility
- U.S. banks offer higher return potential but greater downside risk
- Tail risk is significantly higher for U.S. institutions
- Interest rates and inflation are key macroeconomic drivers
- Portfolio optimization benefits from cross-market diversification